A Appendix A

A.1 Prompt of Entity Extraction and Recognition

Task: Extract **reactants** (substrates) and **products** (target substance) from the given chemical synthesis question. Ignore supplementary reagents, conditions, or procedural steps.
Output Format: 'reactants': [substrates], 'products': [target]

Example:
Input: 'The goal is to produce the target substance ['A'] using substrates ['B', 'C', 'D']. Determine the required supplementary reagents . . .'
Output: 'reactants': ['B', 'C', 'D'], 'products': ['A']

Instructions:
1. Identify lists explicitly marked as 'substrates' and 'target substance' in the input.
2. Exclude any reagents, solvents, or reaction conditions (e.g., 'required supplementary reagents').
3. Follow the format strictly; no additional text or explanations.

Question:

A.2 Question-answer pair format of dataset

To enhance the model's ability to understand different forms of expression, we diversely rephrased each question, generating four different ways of asking.

Question Style

Style 1: I would like to obtain the reaction product $[products]$. Given the required partial reaction substrates $[substrates]$ please help me supplement the remaining required reaction substrates and provide the reaction pathway as well as the conditions required for the reaction.
Style 2: To synthesize compound $[products]$, substrates $[substrates]$ have been provided as initial materials. Please determine the additional substrates needed to complete the reaction sequence, along with the detailed mechanistic pathway and specific experimental conditions required for each step.
Style 3: Given substrates $[substrates]$ as starting materials, propose the missing reagents required for the synthesis of product $[products]$. Include a complete reaction mechanism and provide reaction conditions for each stage.
Style 4: The goal is to produce the target substance $[products]$ using substrates $[substrates]$. Determine the required supplementary reagents for the reaction sequence and provide a step-by-step approach with relevant conditions to achieve optimal synthesis.

Answer Style

The required substrates are $[substrates]$,involving a total of n reaction steps.
Step 1, Using $[A\ and\ B\ and\ \ldots]$ as substrate, $[reagent]$ as reagent. The reaction is carried out at $[temper]$ for $[time]$, which ultimately yields product $[product]$, the reaction equation is $[reaction\ equation]$.
Step 2 is divided into two stages, using $[substrates]$ as substrate, Stage 1,$[solvent]$ as solvent, $[reagent]$ as reagent. The reaction is carried out at $[temper]$ for $[time]$ and at $[atmosphere]$. Stage 2, The reaction is carried out at $[atmosphere]$ for $[time]$ and at $[atmosphere]$, ultimately yielding product $product$, the reaction equation is $[reaction\ equation]$.
. . .

A.3 Knowledge graph triples structure

All substance names are in SMILES format.

Head node: $[head\ node\ name]$
Relationship:{
"condition": "Using $[substrates]$ as substrates, $[solvent]$ as solvent, $[reagent]$ as reagent, $[catalyst]$ as catalyst . . .. The reaction is carried out at $[temper]$ for $[time]$, under $[atmosphere]$. Ultimately yielding product $[product]$, the reaction equation is $[reaction\ equation]$",
"history": "This reaction has been used to generate $[product\ list]$" }
Tail node: $[tail\ node\ name]$

A.4 Formula derivation for based on mutual information retrieval

We can set the optimization objective as maximizing the mutual information between the retrieved subgraph G_s and the user query Q. This optimization objective can be expressed as the following formula.

$$\max_{G_S} I(Q; G_S) = H(Q) - H(Q \mid G = G_S) \tag{5}$$

Here, H represents the Shannon entropy. Since $H(Q)$ is a constant, we can transform the optimization goal into the following form.

$$\begin{aligned} &\min_{G_S} H(Q \mid G = G_S) \\ &= \max_{G_S}\ p(Q \mid G = G_S) \log_2 p(Q \mid G = G_S) \\ &= \max_{G_S} \frac{p(Q, G = G_S)}{p(G = G_S)} \log_2 \frac{p(Q, G = G_S)}{p(G = G_S)} \end{aligned} \tag{6}$$

Thus, the optimization objective is further transformed into the calculation of $\frac{p(Q,G=G_S)}{p(G=G_S)}$. It is worth noting that, in the structure of our constructed knowledge graph, the nodes represent only reactants or products. Therefore, the resulting

subgraph forms a linear, end-to-end chain rather than a divergent graph with many branches. In this structure, $\frac{p(Q,G=G_S)}{p(G=G_S)}$ can be further transformed into the following form.

$$\begin{aligned}
&\frac{p(Q, G=G_S)}{p(G=G_S)} \\
&= \frac{p(Q, h_1, r_1, t_1, h_2, r_2, t_2, \ldots, h_n, r_n, t_n)}{p(h_1, r_1, t_1, h_2, r_2, t_2, \ldots, h_n, r_n, t_n)} \\
&= \frac{p(r_1, t_1, h_2, r_2, t_2, \ldots, h_n, r_n, t_n \mid Q, h_1)}{p(r_1, t_1, h_2, r_2, t_2, \ldots, h_n, r_n, t_n \mid h_1)} \cdot \\
&\frac{p(Q, h_1)}{p(h_1)}
\end{aligned} \tag{7}$$

Where h represents the head node, r represents the relationship, t represents the tail node, and n represents the maximum step count, which will be a hyperparameter. Since question Q and node h_1 are already given in the initial retrieval stage, $p(Q, h_1)$ and $p(h_1)$ are constants, which we ignore. For $p(r_1, t_1, h_2, r_2, t_2, \ldots, h_n, r_n, t_n \mid Q, h_1)$, we can make the following transformation using the conditional probability formula.

$$\begin{aligned}
&p(r_1, t_1, h_2, r_2, t_2, \ldots, h_n, r_n, t_n \mid Q, h_1) \\
&= p(t_1, h_2, r_2, t_2, \ldots, h_n, r_n, t_n \mid Q, h_1, r_1) \cdot \\
&\quad p(r_1 \mid Q, h_1) \\
&= p(h_2, r_2, t_2, \ldots, h_n, r_n, t_n \mid Q, h_1, r_1, t_1) \cdot \\
&\quad p(t_1 \mid Q, h_1, r_1) \cdot p(r_1 \mid Q, h_1) \\
&= p(r_2, t_2, \ldots, h_n, r_n, t_n \mid Q, h_1, r_1, t_1, h_2) \cdot \\
&\quad p(h_2 \mid Q, h_1, r_1, t_1) \cdot p(t_1 \mid Q, h_1, r_1) \cdot \\
&\quad p(r_1 \mid Q, h_1)
\end{aligned} \tag{8}$$

When both reactants and reaction conditions are determined, the products are also determined, which makes $p(t_n|Q, h_n, r_n) = 1$. In addition, in a chain structure, the tail node of the current triplet is also the head node of the next triplet, which makes $p(h_n|Q, h_{n-1}, r_{n-1}, t_{n-1}) = 1$. Based on this, $p(r_1, t_1, h_2, r_2, t_2, \ldots, h_n, r_n, t_n \mid Q, h_1)$ can be transformed as follows.

$$\begin{aligned}
&p(r_1, t_1, h_2, r_2, t_2, \ldots, h_n, r_n, t_n \mid Q, h_1) \\
&= p(t_1, h_2, r_2, t_2, \ldots, h_n, r_n, t_n \mid Q, h_1, r_1) \cdot \\
&\quad p(r_1 \mid Q, h_1) \\
&= p(r_n \mid Q, h_1, r_1, \ldots, t_{n-1}, h_n) \cdot \\
&\quad p(r_{n-1} \mid Q, h_1, r_1, \ldots, t_{n-1}, h_{n-1}) \cdot \\
&\quad \ldots \\
&\quad p(r_2 \mid Q, h_1, r_1, t_1, h_2) \cdot \\
&\quad p(r_1 \mid Q, h_1)
\end{aligned} \tag{9}$$

Similarly, $p(r_1, t_1, h_2, r_2, t_2, \ldots, h_n, r_n, t_n \mid h_1)$ can undergo the same transformation, resulting in the following form.

$$\begin{aligned}
&p(r_1, t_1, h_2, r_2, t_2, \ldots, h_n, r_n, t_n \mid h_1) \\
&= p(t_1, h_2, r_2, t_2, \ldots, h_n, r_n, t_n \mid h_1, r_1) \cdot p(r_1 \mid h_1) \\
&= p(r_n \mid h_1, r_1, \ldots, t_{n-1}, h_n) \cdot \\
&\quad p(r_{n-1} \mid h_1, r_1, \ldots, t_{n-1}, h_{n-1}) \cdot \\
&\quad \ldots \\
&\quad p(r_2 \mid h_1, r_1, t_1, h_2) \cdot \\
&\quad p(r_1 \mid h_1)
\end{aligned} \tag{10}$$

Clearly, the forms of Formula 9 and Formula 10 can be solved through iterative computation. In each iteration, we only need to calculate the conditional probability p of the relationship information between the current known data and the next triplet. Due to the pre-trained nature of LLMs, they inherently possess strong predictive capabilities for the next word. Therefore, we can utilize LLM to compute the conditional probability p, which refers to the probability of the text corresponding to the relation in the next triple appearing, given the existing information ($Q, h_1, \ldots, h_n$ or $h_1, \ldots, h_n$).

A.5 Experimental parameter setting

The training and inference of the model were conducted using an NVIDIA-A40 graphics card with 48GB of VRAM. The specific training and model parameters are shown in Table 5.

Parameter Name	Parameter value
Training hyperparameters	
seed	3
learning_rate	1e-5
weight_decay	0.05
patience	5
train_batch_size	1
grad_steps	2
num_epochs	5
warmup_epochs	1
eval_batch_size	8
max_txt_len	1,600
max_new_tokens	2,048
beam_width	3
jump_num	5
GNN hyperparameters	
gnn_num_layers	4
gnn_in_dim	1,024
gnn_hidden_dim	1,024
alignment_mlp_layers	3
gnn_num_heads	4
distance_operator	euclidean
gnn_dropout	0

Table 5: Experimental parameter setting.

Prompt

You are a chemist, and now you have a task to evaluate the retrosynthetic path of an organic reaction. The evaluation criteria are as follows:

Evaluation Dimensions and Explanations (Total: 100 points)

1. Accuracy of Reactants and Products (30 points total)

This evaluates whether the generated path accurately uses all given starting materials and results in the correct target product, including the scientific rationality of intermediates.

- **Completeness (10 points)**: Does the path include all the specified starting materials? Deduct 3 points for each missing material.
- **Product Accuracy (10 points)**: Is the final product identical to the target product? Deduct 5-10 points for structural errors or functional group mismatches.
- **Intermediate Rationality (10 points)**: Do the intermediates in the reaction conform to organic chemistry rules (e.g., stability of free radicals, carbocations)? Deduct 5-10 points for unreasonable structures.

2. Rationality of the Synthetic Path (40 points total)

This examines the scientific nature, process rationality, and economic feasibility of the overall synthetic path.

- **Reaction Feasibility (20 points)**: Are the reactions in the path reasonable? Do they conform to thermodynamic and kinetic principles? Are there any high-risk or polluting reactions? Deduct 5-10 points for each reaction that doesn't follow basic mechanisms or is unsafe.
- **Step Logic (15 points)**: Does the path follow conventional synthetic sequences (e.g., protection/deprotection, functional group transformation order)? Deduct 1-5 points for redundant steps or disordered sequences.
- **Economic Consideration (5 points)**: Does the path avoid expensive or special reagents as much as possible? Is it industrially feasible? Deduct 1-3 points for the use of rare metals or complex equipment.

3. Rationality of Reaction Conditions (25 points total)

This examines the scientific and practical nature of the reaction conditions in the generated path.

- **Solvent and Catalyst Selection (15 points)**: Is the choice of catalyst/auxiliary agent/solvent reasonable? Deduct 3-5 points for each mismatch or error in common conditions.
- **Reaction Conditions (10 points)**: Are the temperature, time, and other conditions scientifically reasonable? Deduct 2-3 points for each abnormal condition.

4. Clarity of Expression (5 points total)

This evaluates whether the path description uses standardized language and is easy to understand and replicate.

- **Terminology Accuracy (3 points)**: Is SMILES used accurately? Deduct 1-2 points for marking errors or vague naming.
- **Logical Coherence (2 points)**: Is each step clearly described? Are there any ambiguities or logical jumps?

The chemical multi-step reaction pathway inference task and designed path to be evaluated are as follows:

Task:[question]

Answer:[answer]

You need to strictly evaluate the given answer with rigorous standards based on the above scoring criteria. Deduct the corresponding points for any imprecise, unscientific, or non-compliant statements, and output the final score along with the justification.

The final score should be displayed in the format: "**–Final Score: xx points–**"

Table 6: Scoring prompt template.

Model Name	Parameters	Scoring Model	2 Steps	3 Steps	4 Steps	5 Steps	Total Score	Average
LLaMA3.1	8B	GPT4-o1	57.39	56.45	54.61	55.11	55.89	37.78
		Deepseek-R1	30.99	30.62	29.48	30.28	30.34	
		Gemini 2.5Pro	28.54	27.25	26.45	26.19	27.10	
Qwen2.5	7B	GPT4-o1	69.41	67.03	66.97	63.35	66.69	43.20
		Deepseek-R1	33.70	32.99	34.60	32.36	33.41	
		Gemini 2.5Pro	31.81	29.91	28.92	27.32	29.49	
Qwen2.5	72B	GPT4-o1	83.76	81.90	78.97	77.94	80.64	54.59
		Deepseek-R1	43.09	42.86	43.43	42.33	42.93	
		Gemini 2.5Pro	44.06	40.09	38.23	38.38	40.19	
Deepseek-V3	671B	GPT4-o1	91.14	89.28	87.63	85.94	88.49	63.92
		Deepseek-R1	51.69	50.08	50.67	50.47	50.73	
		Gemini 2.5Pro	57.90	53.47	50.17	48.65	52.54	
Qwen3	8B	GPT4-o1	79.85	74.56	71.56	68.97	73.73	47.58
		Deepseek-R1	37.30	36.15	35.03	32.54	35.25	
		Gemini 2.5Pro	38.66	33.86	31.95	30.52	33.75	
LLaMA3.1-ChemKGL	8B	GPT4-o1	84.55	83.11	79.63	76.17	80.87	**71.89**
		Deepseek-R1	56.81	58.02	58.59	58.22	57.91	
		Gemini 2.5Pro	78.20	77.96	77.26	74.14	76.89	
Qwen2.5-ChemKGL	7B	GPT4-o1	80.79	79.30	74.32	75.52	77.48	<u>67.78</u>
		Deepseek-R1	52.42	54.46	53.34	53.06	53.32	
		Gemini 2.5Pro	74.85	74.75	72.07	68.54	72.55	

Table 7: Experimental results for 2,000-Sample Test Set. The best results are in **bold** and the second-best are <u>underlined</u>. "Total Score" refers to the average score given by a single evaluation model across all 2,000 test samples for a specific reasoning model, and n steps indicate the average score of subsets grouped by reaction steps (from 2 to 5 steps), with each step containing 500 samples. Average represents the mean of "Total Score" from the three evaluation models and is used to reflect the overall performance.

Case id	Best model	Reason
21	all wrong	Every model starts to make mistakes in the first step and cannot select the best one.
117	model 5	The product of Model 1 was used as a reactant, Model 2 did not use the given reactant, the previous steps of Models 3, 6, 7, 8, and 9 could not react, only the last step could react reasonably but was meaningless, Model 4 could not react, and Model 5's third and fifth steps could not react. However, compared to other models, Model 5 has a higher accuracy rate.
113	model 6	Model 1, 2, 3, and 4 steps are all unreasonable. Model 5 is reasonable on both sides, but the first step reagent is unreasonable and the target product is described as a reaction substrate. Models 6, 7, 8, and 9 have reasonable reaction steps, but models 7, 8, and 9 describe intermediate products as reaction substrates and the steps and reagent states of the reaction process are not as detailed as model 6. Therefore, model 6 is better.
118	model 7	Model 1 did not answer, models 2, 3, and 4 cannot respond, model 5 has two steps, the second step cannot respond, models 6, 8, and 9 lack a given reactant and multiple steps cannot respond, model 7 has five steps, and the last step cannot respond. Relatively speaking, model 5 has the highest accuracy.
82	model 9	Except for Model 9, which is partially reasonable, most other models are basically unreasonable.

Table 8: Reported some cases of blind review results conducted by chemical experts. Models 1–9 correspond to LLaMA3.1 (8B), Qwen2.5 (7B), Qwen2.5 (72B), Qwen3 (8B), DeepSeek-V3 (671B), LLaMA3.1-ChemKGL (8B), Qwen2.5-ChemKGL (7B), LLaMA3.1-REKG-MCTS (8B), and Qwen2.5-REKG-MCTS (7B), respectively. During the blind evaluation, the experts were not informed of the model identities or sources.